

LAB EXPERIMENT RECORD

SET 12 – WEBSITE TRAFFIC ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	6 october 2026
Day	Tuesday

Aim

To analyze website traffic using line charts, bar charts, tables, and an interactive dashboard in RStudio

Dataset

Date	Page Views
2023-01-01	1500
2023-01-02	1600
2023-01-03	1400

PROGRAM 1: Line Chart – Daily Page Views

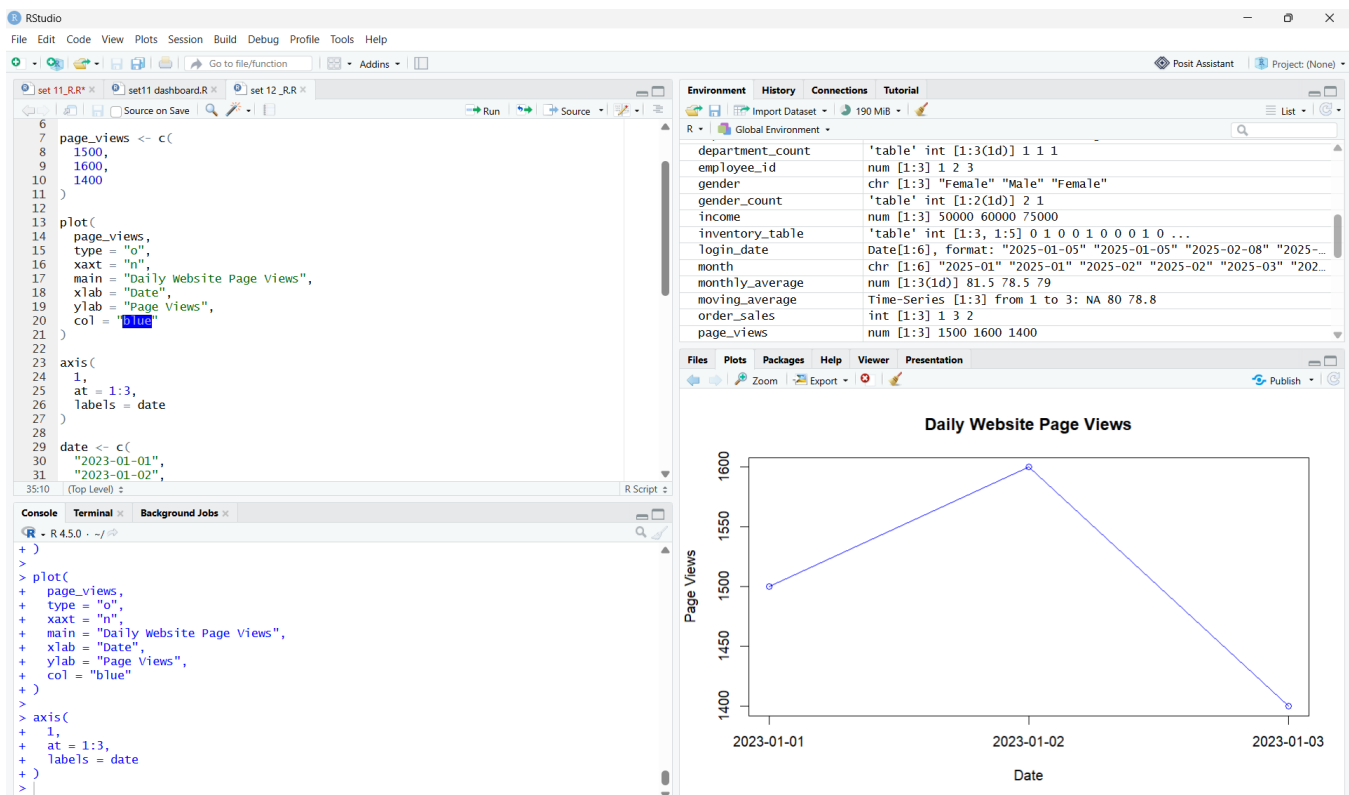
To create a line chart to visualize the trend in daily page views over time

```
# date <- c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03"
)
page_views <- c(
  1500,
  1600,
  1400
)
plot(
  page_views,
  type = "o",
  xaxt = "n",
  main = "Daily Website Page Views",
  xlab = "Date",
```

```

ylab = "Page Views",
col = "blue"
)
axis(
  1,
  at = 1:3,
  labels = date
)= 1
)

```



Output 1: Monthly Sales Line Chart

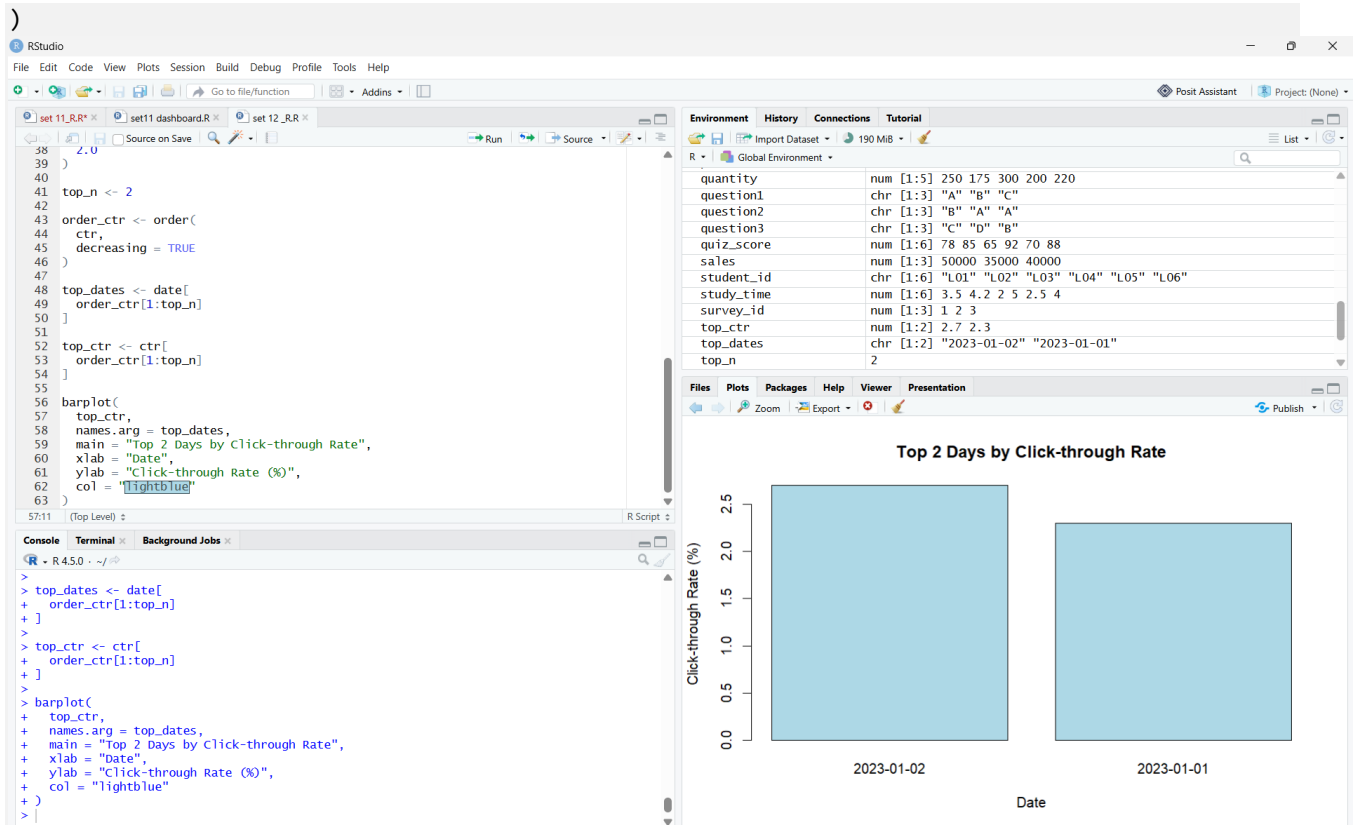
Observation:

Page views increased from 1500 on January 1 to 1600 on January 2, then decreased to 1400 on January 3.

PROGRAM 2: Bar Chart – Top N Days by CTR

To create a bar chart showing the two days with the highest click-through rates.

```
# date <- c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03"
)
ctr <- c(
  2.3,
  2.7,
  2.0
)
top_n <- 2
order_ctr <- order(
  ctr,
  decreasing = TRUE
)
top_dates <- date[
  order_ctr[1:top_n]
]
top_ctr <- ctr[
  order_ctr[1:top_n]
]
barplot(
  top_ctr,
  names.arg = top_dates,
  main = "Top 2 Days by Click-through Rate",
  xlab = "Date",
  ylab = "Click-through Rate (%)",
  col = "lightblue"
```



Observation: January 2 has the highest click-through rate of **2.7%**, followed by January 1 with **2.3%**

PROGRAM 3: Website Traffic Table

To create a table displaying the website traffic information for each day.

```
# date <- c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03")
page_views <- c(
  1500,
  1600,
  1400)
ctr <- c(
  2.3,
  2.7,
  2.0)
traffic_table <- data.frame(
  Date = date,
  Page_Views = page_views,
  Click_Through_Rate = ctr)
traffic_table
```

```
64
65 date <- c(
66   "2023-01-01",
67   "2023-01-02",
68   "2023-01-03"
69 )
70
71 page_views <- c(
72   1500,
73   1600,
74   1400
75 )
76
77 ctr <- c(
78   2.3,
79   2.7,
80   2.0
81 )
82
83 traffic_table <- data.frame([
84   Date = date,
85   Page_Views = page_views,
86   Click_Through_Rate = ctr
87 ])
88
89 traffic_table
```

83:29 (Top Level) R Script

Console Terminal Background Jobs

```
> ctr <- c(
+   2.3,
+   2.7,
+   2.0
+ )
>
> traffic_table <- data.frame(
+   Date = date,
+   Page_Views = page_views,
+   Click_Through_Rate = ctr
+ )
>
> traffic_table
      Date Page_Views Click_Through_Rate
1 2023-01-01      1500                2.3
2 2023-01-02      1600                2.7
3 2023-01-03      1400                2.0
> |
```

Observation

The table displays the daily page views and click-through rates for all three dates

PROGRAM 4: Interactive Dashboard in RStudio

To develop an interactive dashboard combining the daily page-view line chart, CTR bar chart, and website traffic table.

```
library(shiny)

date <- c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03")

page_views <- c(
  1500,
  1600,
  1400)

ctr <- c(
  2.3,
  2.7,
  2.0)

ui <- fluidPage(
  titlePanel(
    "Website Traffic Dashboard" ),
  selectInput(
    "view",
    "Select View:",
    choices = c(
      "Daily Page Views",
      "Top 2 CTR Days",
      "Traffic Table")
  ),
  plotOutput("myplot"),
  tableOutput("mytable")
)

server <- function(input, output) {
  output$myplot <- renderPlot({
    if (input$view == "Daily Page Views") {

      plot(
        page_views,
```

```

    type = "o",
    xaxt = "n",
    main = "Daily Website Page Views",
    xlab = "Date",
    ylab = "Page Views",
    col = "blue"  )
axis(
  1,
  at = 1:3,
  labels = date
) }
if (input$view == "Top 2 CTR Days") {
  order_ctr <- order(
    ctr,
    decreasing = TRUE
  )
  top_dates <- date[
    order_ctr[1:2]
  ]
  top_ctr <- ctr[
    order_ctr[1:2]
  ]
  barplot(
    top_ctr,
    names.arg = top_dates,
    main = "Top 2 Days by Click-through Rate",
    xlab = "Date",
    ylab = "Click-through Rate (%)",
    col = "lightblue"
  )
}
})
output$mytable <- renderTable({
  if (input$view == "Traffic Table") {

    data.frame(

```

```

    Date = date,
    Page_Views = page_views,
    Click_Through_Rate = ctr
  )
}
})
}

```

```

shinyApp(
  ui = ui,
  server = server
)

```

The screenshot shows the RStudio interface with the following components:

- Code Editor:** Contains R code for a Shiny application. The code defines a data frame with columns 'Date', 'Page_Views', and 'Click_Through_Rate'. It also defines the UI and server logic for the dashboard.
- Shiny Preview Window:** Displays the 'Website Traffic Dashboard'. It features a 'Select View:' dropdown menu currently set to 'Traffic Table'. Below the menu is a table showing website traffic data.
- Console:** Shows the execution of the R code, including the creation of the data frame and the Shiny application.

Website Traffic Dashboard

Select View:

Date	Page_Views	Click_Through_Rate
2023-01-01	1500.00	2.30
2023-01-02	1600.00	2.70
2023-01-03	1400.00	2.00

Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard provides three views of product category sales: funnel-style comparison, sales distribution, and detailed sales table.

Procedure:

- Create vectors for product categories and sales.
- Sort the sales values and use a horizontal bar chart as a funnel-style visualization.
- Use `data.frame()` to create the sales table.
- Use `pie()` to display the distribution of sales.
- Use Shiny to combine the funnel-style chart, pie chart, and table.
- Save each question as a separate R script.

Result:

The product category sales data was successfully analyzed using a funnel-style chart, pie chart, and table. An interactive dashboard was also developed in RStudio to explore the product category sales